

F.Y. M.C.A. (Management) CCM-560 MJ : CLOUD COMPUTING MANAGEMENT AND SECURITY (2024 Pattern) (Semester - II)

Question Paper Solution May/April 2025

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.
- 2) All questions carry equal marks.

Question 1(5 Marks Each) (Chapter 1)

A. Explain the six R's of cloud migration with example

Answer:

SIX R's OF CLOUD MIGRATION

Cloud migration is the process of moving data, applications, or other business elements from one place to the cloud.

R	STRATEGY	WHAT IT MEANS	EXAMPLE
1	Re-host (Lift and Shift)	Move the application to the cloud exactly as it is. No changes are made.	Move your old Windows server to an Amazon EC2 or Azure VM.
2	Re-platform (Lift, Tinker, and Shift)	Move the application and make some small improvements to use cloud features. Core code stays same.	Move your database to Amazon RDS or Azure SQL Database.
3	Repurchase (Drop and Shop)	Stop using old software and buy a new cloud software (SaaS). You get new features and automatic updates.	Use Microsoft 365 or Google Workspace instead of your old email server.
4	Refactor / Re-architect (Rewrite)	Rebuild the application to be cloud-native. Better performance, scalable and modern.	Break a big app into small parts (microservices) and run on Kubernetes (EKS / GKE).
5	Retire (Decommission)	Turn off applications that are not needed anymore. Save cost and remove clutter.	Stop an old expense tool that nobody uses. Close the server.
6	Retain (Keep)	Keep some applications on-premises for now. Move them later.	Bank keeps core system in its own data center due to rules, others moved to cloud.

Cloud migration is the process of moving data, applications, or other business elements from one place to the cloud. This could mean moving from:

- **On-premises (local) computers or data centers** to a cloud platform like AWS, Google Cloud, or Azure.
- **One cloud platform to another** (cloud-to-cloud migration).

Six R's of Cloud Migration

1. Re-host (Lift and Shift)

- You pick up your application exactly as it is and move it to the cloud with no changes. Like moving your furniture from one house to another without rebuilding it. This is the fastest migration strategy because you don't rewrite any code. It's often used for quick disaster recovery or to shut down a physical data center urgently.
- Example: A company takes its old Windows server running in the basement and moves it to an Amazon EC2 (AWS) or Azure VM instance exactly as-is.

2. Re-platform (Lift, Tinker, and Shift)

- You move the application but make a few small improvements to take advantage of the cloud, without changing the core code. Like moving your furniture but upgrading the old light bulbs to smart LEDs. You get some cloud benefits without the cost of a full rewrite. This is a good middle-ground option to improve performance without heavy investment.
- Example: Moving a database from a physical server to a managed service like Amazon RDS or Azure SQL Database so the cloud provider handles backups and updates automatically.

3. Repurchase (Drop and Shop)

- You stop using your old software and buy a new one that is already in the cloud. Like throwing away your old landline phone and buying a new smartphone with a plan. You switch to a Software-as-a-Service (SaaS) product instead of maintaining your own. This often includes modern features and automatic updates you didn't have before.
- Example: Ditching an old on-premise email server and buying Microsoft 365 or Google Workspace.

4. Refactor / Re-architect (Rewrite)

- You rebuild the application from scratch to be "cloud-native," making it better, faster, and more scalable. Like tearing down an old house and building a modern smart home designed for how you live now. This takes the most time and money but gives you the best long-term performance. It allows your app to handle millions of users automatically using cloud features.
- Example: Breaking a big monolithic app into small pieces (microservices) running in

containers on Amazon EKS (Kubernetes) or Google Kubernetes Engine.

5. Retire (Decommission)

- You turn off applications you don't need anymore. Why move junk to the cloud? This saves money because you stop paying for electricity, maintenance, and licensing on useless software. It also reduces the clutter in your IT environment, making everything easier to manage.
- Example: A company realizes no one has used their old expense reporting tool in 2 years. Instead of migrating it, they just turn off the server and throw it away.

6. Retain (Keep)

- You leave some applications where they are (on-premises) for now. Maybe they are too sensitive, too expensive to move, or not ready yet. You plan to revisit them later in a future phase of migration. This creates a "hybrid cloud" environment where some things live in the cloud and some stay at home.
- Example: A bank keeps its core transaction processing system in its own data center due to strict government regulations, while moving everything else to the cloud.

B. Discuss the challenges in cloud computing security and propose possible solutions. (Chapter 1)

Answer

Cloud Computing Security Challenges.

- **Data Breaches:** This happens when someone gains unauthorized access to data stored in the cloud. It could be due to weak passwords, poor security settings, or hacking attempts. Protecting sensitive information is crucial to prevent leaks.
- **Data Loss:** Sometimes, data can be accidentally deleted, corrupted, or lost due to technical issues or cyberattacks. If backups are not regularly maintained, lost data might be impossible to recover.
- **Insider Threats:** Not all threats come from outside — sometimes, employees or people with access to cloud systems may misuse their permissions, either intentionally or by mistake, which can put data at risk.
- **Insecure Interfaces and APIs:** Cloud services use APIs (Application Programming Interfaces) to communicate between systems. If these APIs are not properly secured, hackers can exploit them to steal or change data.

- **Account Hijacking:** If someone steals login credentials — through phishing, weak passwords, or other means — they can control cloud accounts, change settings, and access confidential data.
- **Misconfigurations:** Incorrect cloud setup — like leaving storage open to the public or allowing too much user access — can expose data to unauthorized users. Proper configuration is key to security.

Solution

1. **Data Encryption:** Encryption converts data into unreadable form so only authorized users can access it. Even if hackers steal data, they cannot understand it without the key.
2. **Access Control:** Access control ensures only authorized users can enter systems using passwords and MFA. It limits data access based on user roles and permissions.
3. **Regular Security Audits:** Security audits check systems for weaknesses and misconfigurations. They help fix issues early and keep security measures updated.
4. **Backup and Recovery:** Regular backups protect data from loss due to attacks or failures. Recovery systems help restore data quickly and maintain operations.
5. **Firewall and Network Security:** Firewalls monitor and control incoming and outgoing network traffic. They block unauthorized access and protect systems from cyber threats.

C. Discuss the importance of Role based access control (RBAC) in cloud security (Chapter 1)

Answer

Role-Based Access Control (RBAC) is a security method where users are given access based on their job role. Instead of giving full access to everyone, permissions are assigned according to what a person needs to do. It helps organizations manage who can access what in a simple and secure way.

Importance of RBAC:

1. **Improves Security:** Users get only required access, so chances of data misuse or hacking are reduced. This minimizes the risk of unauthorized access to sensitive information.
2. **Follows Least Privilege Principle:** Employees can access only the data needed for their work, not everything. This ensures better control over system resources.
3. **Reduces Human Errors:** Limits accidental deletion or modification of important data. It

prevents users from making changes outside their responsibilities.

4. **Easy Management:** Admin can assign roles instead of managing permissions for each user separately. This saves time and simplifies user management.
5. **Protects Sensitive Data:** Important information is accessible only to authorized roles. This keeps confidential data safe from unauthorized users.

RBAC is important in cloud security because it controls access, protects data, and makes system management simple and secure.

D. Define quality of services (QoS) cloud computing and its significance.(Chapter 1)

Answer

Quality of Service (QoS) refers to the performance and reliability that a cloud provider guarantees to users. It ensures that cloud services run smoothly, efficiently, and meet user expectations.

Key Components of QoS:

1. **Availability (Uptime):** It shows how often the service is available and working. Higher uptime (like 99.9%) means the service is rarely down.
2. **Reliability:** It means the system works properly without failure or data loss. Users can trust that their data will be safe and accessible.
3. **Performance (Speed):** It refers to how fast the service responds to user requests. Good performance reduces waiting time and improves user experience.
4. **Scalability:** It is the ability to handle increased workload by adding resources. It ensures the system works well even when many users access it.
5. **Elasticity:** It means adding or removing resources based on demand. It helps in saving cost by using only required resources.
6. **Security:** It protects data from unauthorized access and cyberattacks. Strong security builds user trust in cloud services.
7. **Cost Efficiency:** It means providing good service at a reasonable cost. Users pay only for what they use, reducing unnecessary expenses.

Significance of QoS:

- **Ensures High Availability:** QoS guarantees that cloud services are available most of the time with minimal downtime.

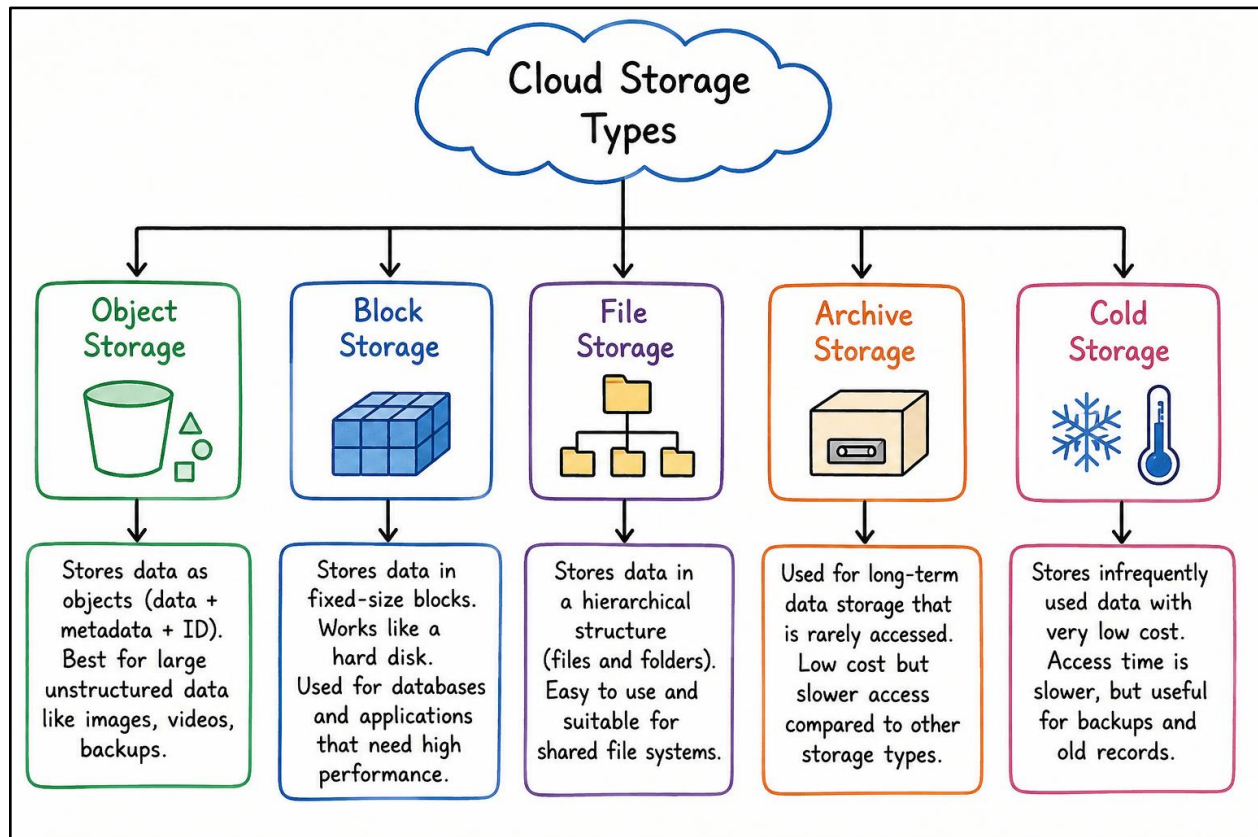
- **Improves Performance:** It ensures fast response time and smooth working of applications.
- **Builds Reliability:** Users can trust the system to work correctly without failures or data loss.
- **Enhances User Satisfaction:** Better service quality leads to a good user experience.
- **Supports Scalability and Flexibility:** QoS helps systems handle increasing users and workload efficiently.
- **Ensures Security:** Protects data and systems from unauthorized access and cyber threats.
- **Cost Efficiency:** Provides value for money by delivering good performance at a reasonable cost.



Question 2 (5 Marks Each)(Chapter 2)

A. What are different storage types in cloud computing? (Chapter 2)

Answer



Cloud computing provides different types of storage to store and manage data efficiently.

1. Object Storage : Stores data as objects (files + metadata + unique ID). Best for storing large unstructured data like images, videos, and backups. It is highly scalable and allows easy access from anywhere.

Example: Used by apps like WhatsApp and Instagram to store photos and videos.

2. Block Storage: Stores data in fixed-size blocks and works like a hard disk. Used for databases and applications that need fast performance. It provides high speed and low latency for critical applications.

Example: Used in banking systems and online transaction databases.

3. File Storage: Stores data in a hierarchical structure (files and folders). Easy to use and suitable for shared file systems. It is commonly used in organizations for file sharing and collaboration.

Example: Used in Google Drive and Dropbox for storing and sharing files.

4. Archive Storage : Used for long-term data storage that is rarely accessed. Low cost but slower access compared to other storage types. Ideal for storing old records and compliance data.
Example: Used by banks or government offices to store old records and documents.

5. Cold Storage: Stores infrequently used data with very low cost. Access time is slower, but useful for backups and old records. It helps reduce storage costs for rarely used data.
Example: Used for backup data in companies or old photos stored in cloud accounts.

B. Compare relational and NoSQL cloud databases with examples.

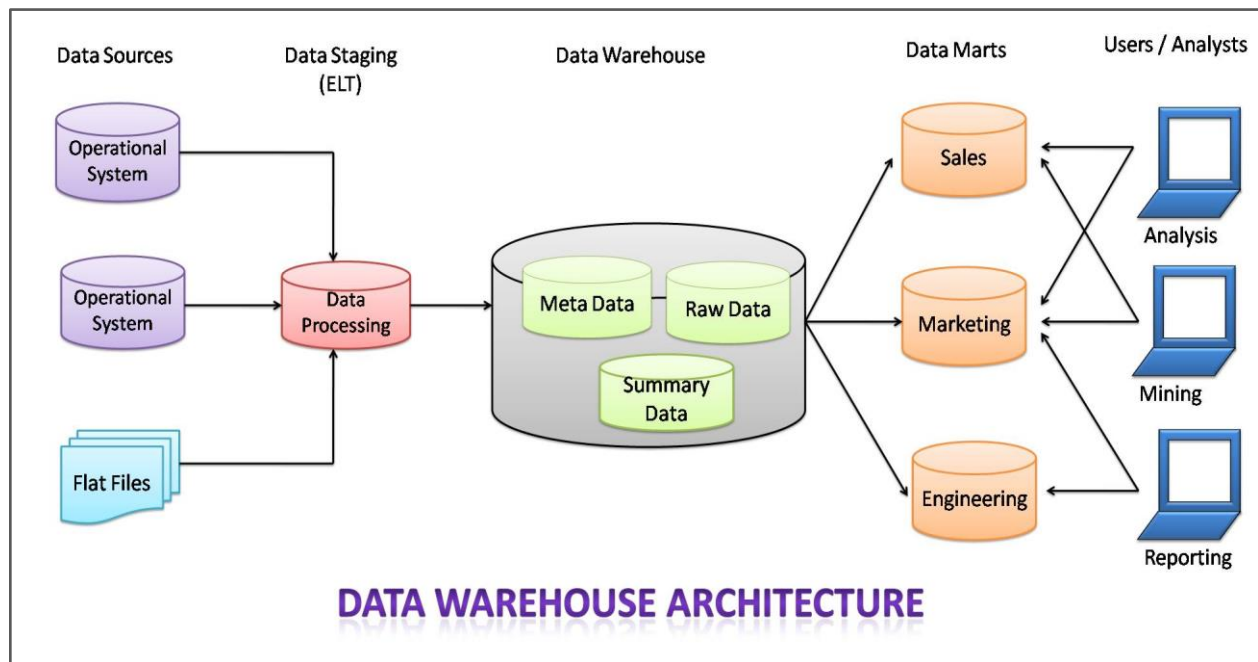
Answer

Relational Database (SQL)	Non-Relational Database (NoSQL)
It stores data in tables with rows and columns (like a spreadsheet).	It stores data in documents, key-value pairs, graphs, or wide columns (like a JSON file).
It has a fixed schema, meaning you must define the structure (columns) before adding data.	It has a dynamic schema, meaning each record can have different fields without prior definition.
It is designed to scale vertically, which means upgrading to a bigger, more powerful server.	It is designed to scale horizontally, which means adding more cheap servers to distribute the load.
It uses SQL (Structured Query Language) to interact with data.	It uses APIs or Unstructured Query Languages (like MongoDB Query Language).
It follows ACID properties (Atomicity, Consistency, Isolation, Durability) to ensure reliable transactions.	It follows BASE properties (Basically Available, Soft state, Eventual consistency) for flexibility and performance.
It is best for complex queries, joins, and multi-row transactions.	It is best for large-scale, high-velocity data with simple queries.
It is ideal for applications where data integrity and consistency are critical (e.g., Banking, ERP).	It is ideal for applications where speed and scalability are critical (e.g., Social media, Gaming, IoT).
Examples: MySQL, PostgreSQL, Oracle, SQL Server.	Examples: MongoDB, DynamoDB, Cassandra, Redis, Firebase.
Cloud examples: Amazon RDS, Azure SQL Database, Cloud SQL.	Cloud examples: Amazon DynamoDB, Azure Cosmos DB, Firestore.
It handles relationships between data very well using Foreign Keys and Joins.	It handles relationships poorly; data is often denormalized or nested within the same record.

C. Explain the primary components and architecture of data warehousing

Answer

A **Data Warehouse** is a centralized repository where data from multiple sources is stored, transformed, and organized to support business intelligence (BI) and decision-making processes.



Primary Components of Data Warehouse

1. Data Sources:

Data sources are systems that provide raw data for analysis. This data can come from operational databases, flat files, cloud storage, or external APIs. Examples include CRM systems, ERP software, and social media platforms. These systems collect daily business transaction data that is later used for analysis.

2. ETL (Extract, Transform, Load):

ETL is the process of collecting and preparing data for the data warehouse. Extract means gathering data from different sources. Transform means cleaning, formatting, and removing errors or duplicates. Load means storing the processed data into the data warehouse for future use.

3. Staging Area:

A staging area is a temporary storage space where raw data is kept before entering the data warehouse. In this area, data is cleaned and checked for errors. It ensures that the data is accurate and consistent.

After processing, the data is moved to the warehouse.

4. Data Warehouse Storage:

The data warehouse is the main storage system where integrated and historical data is stored. The data is organized using schemas like Star or Snowflake. It helps in analyzing large amounts of information. It can be stored on-premises or in cloud platforms.

5. Data Marts:

Data marts are smaller sections of a data warehouse created for specific departments such as sales or marketing. They contain only relevant data needed by that department. This makes data access faster and easier for users. It helps teams make better decisions quickly.

6. Metadata:

Metadata means data about data. It describes the structure, source, and meaning of the data stored in the warehouse. Technical metadata includes table names and schema details, while business metadata includes KPI definitions and business rules. It helps users understand the data clearly.

7. OLAP (Online Analytical Processing):

OLAP tools are used to analyze data from different perspectives. They allow users to perform operations like slicing, dicing, drill-down, and roll-up. This makes it easier to study trends and patterns. There are different types of OLAP models such as ROLAP, MOLAP, and HOLAP.

8. Reporting and Visualization:

Reporting and visualization tools help users understand data through dashboards, charts, and reports. These tools present complex data in a simple and clear format. They support better decision-making by showing insights visually. Examples include Power BI and Tableau.

D Compare different cloud database services like Amazon Aurora, Amazon DynamoDB and Amazon Neptune.

Answer

Feature	Amazon Aurora	Amazon DynamoDB	Amazon Neptune
Database Type	Relational (SQL)	NoSQL	Graph Database
Data Model	Tables (rows & columns)	Key-Value / Document	Nodes & Edges
Schema	Fixed schema	Flexible schema	Flexible schema
Query Language	SQL	API / PartiQL	Gremlin / SPARQL
Scalability	Auto-scaling (limited horizontal)	Fully serverless, highly scalable	Scales for relationship queries
Performance	High (optimized for transactions)	Very high (low latency)	High for connected data
Best Use Case	Banking, ERP, transactional apps	Gaming, IoT, real-time apps	Social networks, recommendations
Data Relationships	Strong relationships (joins)	Limited relationships	Complex relationships
Storage Type	Structured data	Semi-structured / unstructured	Highly connected data
Management	Managed service	Fully managed (serverless)	Fully managed
Backup & Recovery	Automatic backups	Built-in backup	Automatic backups
Cost Model	Pay for instances & storage	Pay-per-request	Pay for instances & storage

Question 3 (5 Marks Each)(Chapter 3)

A. Discuss the AWS security services and their role in preventing cyber threats (write Any 5 point in the exam)

Answer

AWS Security Services

- **AWS IAM (Identity and Access Management):** AWS IAM is a service that helps you control who can access AWS resources. It allows you to create users, groups, and roles and assign permissions to them. This ensures that each person gets only the access they need (principle of least privilege).
Example: In a company, a manager can access reports, but a developer can only access code. IAM makes sure employees only see what they are allowed to see.
- **AWS KMS (Key Management Service):** AWS KMS is used to create and manage encryption keys. It helps protect sensitive data by encrypting it, so even if data is stolen, it cannot be read without the key.
Example: A company encrypts customer credit card details before storing them in a database so that even hackers cannot read them.
- **AWS Shield (DDoS Protection):** AWS Shield protects applications from DDoS attacks, where attackers send huge traffic to crash a website. It automatically detects and blocks such attacks.
Example: An online shopping website stays active during a sale even when attackers try to overload it with fake traffic.
- **AWS Detective:** AWS Detective helps in analyzing and investigating security issues. It collects data and uses machine learning to find unusual activities and possible threats.
Example: If someone logs in from another country at an unusual time, AWS Detective helps identify and investigate this suspicious activity.
- **AWS WAF (Web Application Firewall):** AWS WAF protects web applications by filtering and blocking harmful traffic such as SQL injection, cross-site scripting, and bot attacks.
Example: If a hacker tries to enter malicious code into a login form, AWS WAF blocks the request and keeps the website safe.
- **AWS Config:** AWS Config monitors AWS resources and checks whether they are configured according to security rules. It also keeps a history of configuration changes.
Example: If a storage bucket becomes public by mistake, AWS Config alerts the administrator to fix it.

- **AWS CloudTrail:** AWS CloudTrail records all actions taken in an AWS account. It provides a history of user activity and API usage for auditing and security analysis.
Example: If important data is deleted, CloudTrail logs help identify who deleted it and when.
- **Amazon Guard Duty:** Guard Duty is a threat detection service that continuously monitors AWS accounts for suspicious activities, such as unauthorized access or malware.
Example: If someone tries to log in using stolen credentials, Guard Duty alerts the company immediately.
- **AWS Artifacts:** AWS Artifacts provides access to security and compliance reports. It helps organizations prove that they meet legal and industry standards.
Example: A bank uses AWS Artifacts reports to show auditors that their cloud system follows security regulations.
- **AWS Security Hub:** AWS Security Hub gives a central view of all security alerts from different AWS services. It helps in managing and prioritizing risks easily.
Example: Instead of checking multiple tools, a company sees all security warnings in one dashboard and fixes issues quickly.
- **AWS Secrets Manager:** Secrets Manager securely stores and manages sensitive information like passwords, API keys, and database credentials. It also supports automatic rotation of secrets.
Example: A database password is stored securely in Secrets Manager instead of writing it in application code.
- **AWS CloudWatch:** AWS CloudWatch monitors applications and resources in real time. It collects data, tracks performance, and sends alerts if something goes wrong.
Example: If a server's CPU usage becomes too high, CloudWatch sends an alert so the admin can take action before the system crashes.

Role in preventing cyber threats

- Protect sensitive data through encryption and secure storage.
- Control and manage user access to AWS resources.
- Detect and prevent cyberattacks such as hacking and DDoS attacks.
- Monitor system activities and identify suspicious behavior.
- Maintain security compliance and auditing records.
- Provide real-time alerts and centralized security management.
- Ensure availability, confidentiality, and integrity of cloud resources.
- Reduce security risks and improve overall cloud protection.

B. Explain network security in cloud computing

Answer

- Network security is the practice of protecting computers, data, and systems from cyber threats like hackers, viruses, and unauthorized access.
- It ensures that only authorized users can access important information while keeping attackers out. Common security measures include firewalls, encryption, VPNs, and antivirus software.
- With strong network security, businesses and individuals can safely use the internet without the risk of data theft or cyberattacks.
- It plays a crucial role in keeping online activities private and secure.

Key Components of Network Security

1. Firewall

- A firewall is like a security guard for your network. It blocks harmful traffic and only allows safe data to pass through.
- **Example:** A firewall stops hackers from accessing your personal files while allowing you to browse the internet safely.

2. Intrusion Detection & Prevention System (IDPS)

- **Intrusion Detection System (IDS):** Monitors network activity and **alerts** if it detects any suspicious behavior.
- **Intrusion Prevention System (IPS):** **Blocks** threats before they can cause harm.
Example: If someone tries to hack your company's system, the IPS will **detect and stop** the attack.

3. Virtual Private Network (VPN)

- A VPN creates a secure tunnel for sending and receiving data over the internet. It hides your IP address and encrypts your data so hackers cannot see your online activity.
- **Example:** When using public Wi-Fi at a coffee shop, a VPN protects your banking details from hackers.

4. Access Control

- Access control ensures that **only authorized users** can enter a system or network. It checks **who is allowed in** and **what they can do**.

- **Example:** A company ensures that **employees** can access work files, but **outsiders** cannot.

5. Antivirus & Anti-Malware Software

- This software **detects and removes viruses, spyware, ransomware, and other malicious programs** from your computer or network.
- **Example:** If you accidentally download a harmful file, antivirus software **scans and removes** it before it causes damage.

6. Encryption

- Encryption **converts data into a secret code** so that only the intended recipient can read it. Even if hackers intercept the data, they cannot understand it.
- **Example:** When you send a **WhatsApp message**, it is **encrypted**, so only the person you sent it to can read it.

7. Network Segmentation

- Dividing a network into smaller sections to **limit damage** if one part gets attacked.
- **Example:** If an attacker breaches one office's Wi-Fi, they **cannot access** other office branches.

C. How does AWS IAM (Identity & Access Management) improve cloud security

Answer

Amazon Web Services IAM (Identity & Access Management) improves cloud security by controlling who can access cloud resources and what actions they can perform. It helps organizations protect data, applications, and services from unauthorized access.

How AWS IAM Improves Cloud Security

1. Identification

- Users must have a unique identity, such as a **username, email ID, or employee ID**.
- Example: When logging into a company's portal, users enter their username.

2. Authentication

- Verifies the user's identity using **passwords, biometrics (fingerprints, face scan), OTPs, or multi-factor authentication (MFA)**.

- Example: A bank's website asks for an OTP sent to your phone after entering your password.

3. Authorization

- After authentication, IAM determines **what resources the user can access** based on assigned permissions.
- Example: A regular employee can view files, but only an admin can modify or delete them.

4. User Provisioning & De-provisioning

- **Provisioning:** Assigning new employees access to necessary applications and systems.
- **De-provisioning:** Removing access when an employee leaves the company.
- Example: A company disables the email and database access of a former employee.

5. Single Sign-On (SSO)

- Allows users to log in **once** and access multiple applications **without entering credentials repeatedly**.
- Example: Logging into Google automatically signs you into Gmail, YouTube, and Drive.

6. Multi-Factor Authentication (MFA)

- Requires users to **verify their identity using two or more methods** (e.g., password + OTP).
- Example: Social media apps ask for a password and then send a verification code to your phone.

7. Role-Based Access Control (RBAC)

- Users are given access **based on their job roles** (e.g., employee, manager, admin).
- Example: A software developer has access to code repositories, but an HR employee does not.

8. Auditing & Monitoring

- Tracks user activities, login attempts, and security events to **detect unauthorized access**.
- Example: A company's security team reviews logs of login attempts to spot suspicious activity.

D. Explain Encryption, Hash functions and VPN as security measures in AWS

Answer

1. Encryption in AWS

Amazon Web Services Encryption is a security technique used to convert readable data (plaintext) into unreadable data (ciphertext) so that unauthorized users cannot access it.

Types of Encryptions in AWS

1. **Symmetric Encryption (Same Key):** In symmetric encryption, one key is used to lock (encrypt) and unlock (decrypt) the data.
2. **Asymmetric Encryption (Two Keys)** In asymmetric encryption, two different keys are used:
 - Public key → for encryption (shared with everyone)
 - Private key → for decryption (kept secret)

AWS Services for Encryption

- AWS KMS (Key Management Service) – Manages encryption keys securely.
- AWS CloudHSM – Provides dedicated hardware security modules for strong encryption.

Example: If a company stores customer information in Amazon S3, encryption ensures that even if data is stolen, it cannot be read without the decryption key.

Benefits

- Protects sensitive information
- Prevents unauthorized access
- Helps meet compliance and security standards

2. Hash Functions in AWS

A hash function converts data into a fixed-length value called a hash or digest. It is mainly used for data integrity and password security.

Features of Hash Functions

- Same input always gives the same output
- One-way process (cannot easily reverse the original data)
- Small data changes create completely different hashes

Common Hash Algorithms

- SHA-256
- SHA-1
- MD5 (older and less secure)

Uses in AWS

- Password verification
- Data integrity checking
- Digital signatures and authentication

Example

When a user logs into an AWS application, the password is compared using its hash value instead of storing the actual password.

Benefits

- Ensures data integrity
- Improves authentication security
- Detects unauthorized changes in files or data

3. VPN (Virtual Private Network) in AWS

A VPN creates a secure and encrypted connection between a user's network and the AWS cloud over the internet.

AWS VPN Services

- AWS Site-to-Site VPN – Connects office networks to AWS securely.
- AWS Client VPN – Allows remote users to securely access AWS resources.

Working of VPN

VPN encrypts all data transferred between the user and AWS, preventing hackers from intercepting sensitive information.

Example

A company employee working from home can securely connect to the company's AWS environment using AWS Client VPN.

Benefits

- Secure remote access
- Encrypted communication
- Protects data from cyber attacks during transmission

Question 4 (Chapter 4)

A. Explain the best practices for AWS Backup & Disaster Recovery

Answer

Backup and Disaster Recovery (DR) are important to keep data safe and continue work even after system failure or disaster.

1. **Follow 3-2-1 Backup Rule:** Keep three copies of data, store them on two different media, and keep one copy offsite (like cloud storage such as Amazon S3). This reduces the risk of losing all data at once. If one copy is damaged, other copies are still available. It is a simple and very effective method for data safety.
2. **Automate Backups:** Use services like AWS Backup to schedule backups automatically. This saves time and reduces human errors. Backups will happen regularly without forgetting. It ensures data is always updated and protected.
3. **Define RTO and RPO:** RTO means how quickly the system should be recovered after failure. RPO means how much data loss is acceptable. These help in planning backup and recovery strategies.
It improves system availability and reduces downtime.
4. **Data Security and Encryption:** Always encrypt backup data to keep it safe. Use proper access control so only authorized users can access it. This protects data from hackers and unauthorized access.
Security is very important for sensitive information.
5. **Regular Testing and Monitoring:** Test backups regularly to check if data can be restored. Monitor the backup process to detect failures early. This ensures backups are working properly. It avoids problems during real disaster situations.
6. **Use Multiple Locations:** Store backups in different locations or regions using Amazon Web Services. If one location fails, data is still safe in another place. This protects against natural disasters and system failures. It increases reliability and availability of data.

B. Describe the AWS backup services and its role in automated database backup

Answer

AWS Backup is a fully managed service provided by Amazon Web Services that helps to **centralize and automate data backups** across AWS resources like databases, storage, and file systems.

AWS Backup Services:

AWS Backup allows users to:

- Create **backup plans** to define when and how backups should be taken
- Manage backups of services like Amazon RDS and Amazon DynamoDB
- Store backups securely in backup vaults
- Monitor and manage backups from a single dashboard

It reduces manual effort and ensures consistent data protection.

Role in Automated Database Backup:

1. Automatic Backup Scheduling:

AWS Backup allows setting schedules (daily, weekly). Backups are taken automatically without manual work. This ensures regular and timely database backups.

2. Centralized Management:

All database backups are controlled from one place. No need to manage each database separately. It simplifies administration and saves time.

3. Data Security and Protection:

Backups are encrypted and securely stored. Access is controlled using permissions. This protects sensitive database information.

4. Point-in-Time Recovery:

For databases like Amazon RDS, data can be restored to a specific time. It helps recover from accidental deletion or errors. This improves reliability.

5. Easy Restore and Disaster Recovery:

Data can be restored quickly when needed. Supports disaster recovery planning. Helps maintain business continuity.

C. A financial services company faced a data breach when their AWS RDS backups were publicly exposed due to misconfiguration of S3 Bucket policies. How can IAM policies and encryption improve backup security (10 Marks) (Chapter 4 & 3)

Answer

Introduction

A healthcare organization lost important patient data because their cloud-hosted database was accidentally deleted. Patient records contain medical history, reports, prescriptions, and treatment details. Loss of this data creates serious problems because hospital work may stop, patient treatment may be delayed, and patient trust may be affected. Since healthcare data is highly sensitive, quick recovery is very important.

Cause of Problem

The main reason for this issue was accidental deletion of the database and the absence of a proper disaster recovery plan. The organization may not have enabled regular backups, automated snapshots, or recovery options like Point-in-Time Recovery (PITR).

Another reason could be poor IAM configuration. If many users have permission to delete or modify the database, the chance of accidental deletion becomes higher. Lack of Multi-Factor Authentication (MFA) and weak monitoring systems also increase the risk.

Without proper logging and alerts, the organization may not know immediately when deletion happens, which delays recovery and increases data loss.

Solution / Proactive Measures

1. Enable Automated Backups

The organization should enable AWS RDS automated backups so that database backups are created regularly without manual work. This helps in restoring lost data easily and reduces the risk of permanent data loss during accidental deletion or system failure.

2. Use Manual Snapshots

Manual snapshots should be taken before major updates or changes in the database. These snapshots help restore the database if accidental deletion happens. They provide an extra safety layer before performing important operations.

3. Enable Point-in-Time Recovery (PITR)

Point-in-Time Recovery allows restoring the database to a specific time before the deletion happened. This reduces data loss and helps quick recovery. It is especially useful when recent transactions need to be recovered accurately.

4. Use AWS Backup

Amazon Web Services AWS Backup service helps manage all backups from one place. It can schedule backups automatically and monitor backup status easily. This improves backup management and ensures regular data protection without manual effort.

5. Apply IAM Security

IAM should be used to control who can delete or modify the database. Only authorized users should get access by following the principle of least privilege. This reduces the chances of accidental deletion by unauthorized users.

6. Enable MFA

Multi-Factor Authentication (MFA) should be enabled for admin users to provide extra security and prevent unauthorized actions. Even if a password is stolen, MFA helps stop unauthorized access.

7. Use Multi-AZ Deployment

Multi-AZ deployment keeps a copy of data in another availability zone. This improves availability and helps faster recovery during failures. It also reduces downtime and keeps hospital services running smoothly.

8. Monitoring with CloudTrail and CloudWatch

AWS CloudTrail records activities like deletion events, and CloudWatch sends alerts for suspicious actions. This helps in quick response and recovery. It also helps administrators identify the cause of problems and take preventive action quickly.

Conclusion

By using proper disaster recovery planning, regular automated backups, manual snapshots, Point-in-Time Recovery, IAM security, MFA, Multi-AZ deployment, and monitoring tools like CloudTrail and CloudWatch, healthcare organizations can protect important patient data effectively.

These AWS tools help reduce the risk of accidental deletion, improve data security, and ensure quick recovery during emergencies. This improves system reliability, patient trust, and smooth hospital operations.

Question 5 (short Note) (Chapter 5)

A. AWS Lambda

Answer

AWS Lambda is a serverless computer service offered by Amazon Web Services (AWS) that allows you to run code without the need to provision, manage, or scale servers manually. With Lambda, you simply upload your code, and the service automatically handles everything required to run and scale it based on demand. This makes it an ideal solution for applications that require flexible scaling, event-driven execution, and cost-efficiency, as you're only charged for the actual compute time your code consumes.

Lambda supports multiple programming languages like Python, Node.js, Java, and more, and integrates seamlessly with other AWS services such as S3, DynamoDB, API Gateway, and CloudWatch. It is widely used in scenarios like real-time file or data processing, backend services for web and mobile apps, and automation tasks. By eliminating the need for server management, AWS Lambda enables developers to focus solely on writing and deploying code quickly, making it a cornerstone of modern, cloud-native application development.

Key Features:

- **Serverless:** No need to manage infrastructure. AWS handles provisioning, scaling, and availability.
- **Event-driven:** Lambda functions can be triggered by AWS services like S3, DynamoDB, API Gateway, or custom events.
- **Automatic scaling:** Lambda automatically scales the number of instances based on the number of incoming requests.
- **Pay-per-use:** You are charged only for the compute time your code uses, in increments of milliseconds.

Use Cases:

- Real-time file processing (e.g., image or video upload to S3)
- Backend for web or mobile apps
- Real-time data streaming using services like Kinesis
- Automation and scheduled tasks (using Amazon Event Bridge or CloudWatch)

How it works:

- You write a small piece of code (called a **function**)

- You set a **trigger** (like file upload, API request, etc.)
- Whenever the event happens, Lambda **runs your code automatically**

Benefits:

- No server management
- Efficient cost model
- High scalability
- Tight integration with AWS ecosystem
-

B. Amazon EC2

Answer

Amazon Elastic Compute Cloud (Amazon EC2) is a cloud service provided by Amazon Web Services (AWS) that offers scalable and flexible virtual servers. These virtual servers, called instances, allow users to run applications in the cloud without having to invest in expensive physical hardware.

Think of Amazon EC2 as a computer rental service on the internet. Instead of buying a physical computer (server), you can rent a virtual one from AWS and use it for as long as you need. You can choose the type of virtual server based on how much power (CPU), memory (RAM), and storage you need.

Example: Imagine you are a software developer and want to **host a website**. Instead of buying a physical server and managing it, you can:

1. **Create an EC2 instance** – This is like turning on a virtual computer.
2. **Install a web server (e.g., Apache, Nginx, or IIS)** – This helps run your website.
3. **Deploy your website files** – You upload and run your website.
4. **Make it available to users** – People can access your site via the internet.

When you don't need the server anymore, you can **stop or delete the EC2 instance**, and you will only be charged for the time you used it.

Key Features of Amazon EC2

- **On-Demand Instances:** You can create instances whenever needed and pay only for the time you use them.
- **Scalability (Auto Scaling) :** EC2 allows automatic scaling, which means it can **increase or**

decrease the number of servers based on demand.

- **Multiple Instance Types:** EC2 provides different types of instances optimized for computing, memory, storage, or networking needs. Examples:
 - **T-Series (General Purpose):** Good for websites and small applications.
 - **C-Series (Compute-Optimized):** Ideal for high-performance applications.
 - **R-Series (Memory-Optimized):** Best for databases and big data applications.
- **Security & Reliability:** EC2 instances are protected by **AWS security features**, including firewalls, identity management, and encryption.
- **Integration with AWS Services:** EC2 works with other AWS services like:
 - **Amazon S3** (Storage)
 - **Amazon RDS** (Databases)
 - **AWS IAM** (Security & Access Management)

Key Benefits of EC2

- **Scalable** – You can increase or decrease resources based on demand.
- **Cost-Effective** – You only pay for what you use (Pay-as-you-go model).
- **Flexible** – You can choose different operating systems like Windows, Linux, etc.
- **Secure** – AWS provides security features to protect your data.

C. Amazon EC2 LightSail

Answer

Amazon LightSail is a cloud service from AWS that makes it easy to launch and manage virtual private servers (VPS). It is designed for students, developers, small businesses, and people who are new to cloud computing. LightSail provides everything you need to build websites and web applications at a low cost. A VPS is a virtual machine (computer) that runs in the cloud. It works like a regular computer but is hosted online. You can install software, host websites, run applications, and store files.

Features of Amazon LightSail:

1. **Easy to Set Up** – Launch a server in just a few clicks.
2. **Pre-configured Apps** – Includes WordPress, Magento, Joomla, etc.

3. **Fixed Pricing** – Simple monthly pricing with no hidden costs.
4. **Built-in Networking** – Includes static IP, DNS management, and firewall.
5. **Snapshots** – Take backups of your server.
6. **SSH Access** – Securely connect to your VPS using terminal/command line.

What Can You Do with LightSail?

- Host websites and blogs.
- Run e-commerce platforms.
- Develop and test applications.
- Host learning platforms or projects.
- Set up file storage or game servers.

Why Use Amazon LightSail?

- Beginner-friendly alternative to Amazon EC2.
- Easy to use compared to other complex cloud services.
- Ideal for students and developers who want to **experiment, learn, or build small projects**.
- Offers a **free trial** for new users.

Example: Launching a WordPress Website with LightSail

1. **Login** to your AWS account and open LightSail.
2. Click "**Create Instance**".
3. Choose **Instance Location** (region).
4. Select "**Linux/Unix**" and choose **WordPress** from the application blueprint.
5. Choose your **instance plan** (e.g., \$3.50/month).
6. Name your instance and click **Create Instance**.
7. After launch, LightSail provides a **public IP**—you can access your website by typing it into a browser.
8. Login to your WordPress admin dashboard using the default credentials provided.

D. Amazon EC2 Auto Scaling

Answer

Amazon EC2 Auto Scaling is a service that **automatically adjusts** the number of EC2 instances in your application based on demand. It ensures that you always have the right number of instances running—**increasing capacity during high traffic** and **reducing instances when demand is low** to save costs.

How It Works:

1. **Monitor Demand** – Auto Scaling tracks the usage of your application (CPU, memory, network traffic, etc.).
2. **Scaling Up (Increase Capacity)** – When more users visit your application, EC2 Auto Scaling adds more instances to handle the load.
3. **Scaling Down (Reduce Capacity)** – When traffic decreases, it removes unnecessary instances to save costs.

Key Benefits:

- **High Availability** – Ensures your application remains responsive even during traffic spikes.
- **Cost Optimization** – Avoids paying for unused capacity by automatically removing idle instances.
- **Automation & Efficiency** – No manual intervention is needed; Auto Scaling works in real-time.

Example :Imagine you run an online store. During normal hours, you have a few customers, but during sales or festive seasons, thousands of people visit your store at the same time. If your website doesn't have enough servers to handle the traffic, it may slow down or crash.

Amazon EC2 Auto Scaling helps solve this problem by automatically adding more servers (instances) when demand increases and reducing them when demand drops.